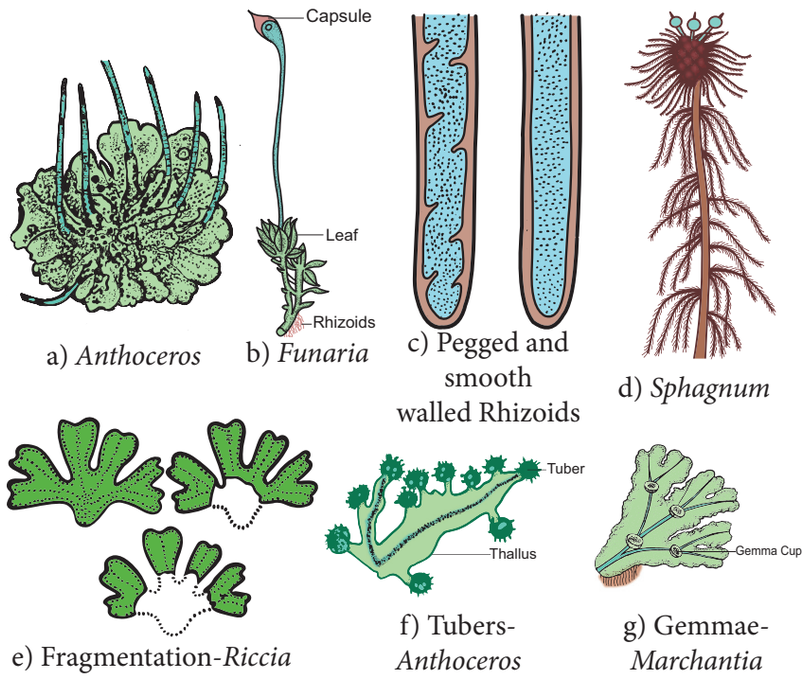


a) *Anthoceros* b) *Funaria* c) Pegged and smooth walled Rhizoids d) *Sphagnum*

e) Fragmentation-*Riccia* f) Tubers-*Anthoceros* g) Gemmae-*Marchantia*

Figure 2.5: Structure and reproduction in Bryophytes

or brood bodies are formed, they help in vegetative reproduction as in *Bryopteris fruticulosa*. In *Marchantia* propagative organs called gemmae are formed and help in reproduction.

- Sexual reproduction is oogamous. Antheridia and Archegonia are produced in a protective covering and are multicellular.
- The antheridia produces biflagellate antherozoids which swims in thin film of water and reach the archegonium and fuse with the egg to form diploid zygote.
- Water is essential for fertilization.
- The zygote is the first cell of the sporophyte generation. It undergoes mitotic division to form multicellular undifferentiated embryo. The embryogeny is exoscopic (the first division of the zygote is transverse and the apex of the embryo develops from the outer cell). The embryo divides and give rise to sporophyte.
- The sporophyte is dependent on gametophyte.
- It is differentiated into three recognizable parts namely foot, seta and capsule.
- Foot is the basal portion and is embedded in the gametophyte through which water and nutrients are supplied for the sporophyte. The diploid spore mother cells found in the capsule region undergoes meiotic division and give rise to haploid spores. Bryophytes are homosporous. In some sporophytes elaters are present and help in dispersal of spores (Example: *Marchantia*). The spores germinate to produce gametophyte.
- The zygote, embryo and the sporogonium constitute sporophytic phase. The green long living haploid phase is called gametophytic phase. The haploid gametophytic phase alternates with diploid sporophyte and shows heterologous alternation of generation.

Proskauer in the year 1957 classified Bryophytes into 3 Classes namely

i. **Hepaticopsida** (*Riccia, Marchantia, Porella* and *Riella*)

ii. **Anthocerotopsida** (*Anthoceros* and *Dendroceros*)

iii. **Bryopsida** (*Funaria, Polytrichum* and *Sphagnum*).

2.4.2 Economic importance

Dead thalli of *Sphagnum* gets accumulated and compressed, hardened to form peat. In northern Europe (Netherlands)peat is used as fuel in commercial scale. Apart from this nitrates, brown dye and tanning materials are derived from peat. *Sphagnum*

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